CYBERSECURITY LAB 08

Wazuh SOC Lab Build, Validation & Troubleshooting Manual

Phases 1-7 | VMware ESXi, Ubuntu, Windows 11, Wazuh SIEM

Version 1.0 | November 2, 2026

Evidence policy: This manual distinguishes confirmed lab observations from reproducible reference commands. Some early-phase installation command transcripts were not available in the recovered session record. Those procedures are identified as reference/reconstruction, not independently verified execution.

Safety: All authentication tests described were authorized and confined to the lab. Do not publish credentials, agent keys, private certificates, or full sensitive logs in a public GitHub repository.

1. Executive overview and evidence legend

Objective: build a working SOC monitoring environment, enroll a Windows endpoint, verify Windows event ingestion and file-integrity monitoring, assess security configuration, perform a controlled authentication test, investigate the resulting alert, and document incident closure.

Mark	Meaning
VERIFIED	Observed directly in user-provided PowerShell output, screenshots, or confirmed lab results.
RECONSTRUCTED	Supported by the known configuration or earlier conversation summary, but the full original command transcript is not available.
REFERENCE	Recommended command to reproduce/check a configuration; not proof that the command was originally executed.

Phase completion overview

Phase	Topic	Recorded outcome
1	SOC VM and infrastructure	Ubuntu SOC VM and network configuration verified
2	Wazuh server stack	Wazuh 4.14.9 manager, indexer, dashboard and Filebeat active
3	Windows agent and log ingestion	IAM-MGMT01 agent 001 active; Application Event 101 observed
4	File integrity monitoring	Create/modify/delete alerts: rules 554/550/553
5	Security posture and authentication monitoring	SCA 26%; 4624 and 4625 monitoring verified; no AD policy changes
6	Controlled security event generation	Local failed logon Event 4625 / Record 46309; cleanup verified
7	SOC investigation and closure	Incident LAB08-SOC-003 documented; five evidence files present

2. Architecture and network inventory

Verified topology: VMware ESXi 8 hosts an Ubuntu SOC VM. A Windows 11 domain-joined endpoint sends Wazuh agent telemetry to the SOC server over TCP 1514. The dashboard is reached over HTTPS.

```
IAM-MGMT01 (Windows 11 Pro)              soc-ubuntu01 (Ubuntu 24.04.3)
192.168.150.30                             192.168.150.50
Wazuh Agent 4.14.9  -- TCP 1514 -->       Wazuh Manager 4.14.9
Windows Event Logs                         Wazuh Indexer 4.14.9
FIM: C:\SOC-Lab08                         Wazuh Dashboard 4.14.9
                                           Filebeat 7.10.2
                 Browser -- HTTPS 443 --> Dashboard
```

Component	Value / verification
Hypervisor	VMware ESXi 8
SOC host	soc-ubuntu01, Ubuntu 24.04.3, 192.168.150.50/24
SOC gateway / DNS	192.168.150.1 / 192.168.150.10

Component	Value / verification
SOC resources	Approximately 11 GiB RAM and 145 GiB disk after LVM expansion
Windows endpoint	IAM-MGMT01, Windows 11 Pro, 192.168.150.30
Domain	corp.example.test
Agent identity	001; Wazuh agent 4.14.9; default group
Dashboard	https://192.168.150.50 (self-signed certificate warning)

Boundary: This is the Lab 08 192.168.150.0/24 architecture. It must not be confused with separate historical examples using 192.168.0.0/24 or unrelated network-monitoring products.

3. Phase 1 - Ubuntu SOC virtual machine

Known build state - VERIFIED / RECONSTRUCTED

• Ubuntu 24.04.3 SOC VM on ESXi 8, address 192.168.150.50/24, default gateway 192.168.150.1, DNS 192.168.150.10.

• Disk capacity expanded through LVM to approximately 145 GiB; approximately 11 GiB RAM available.

Repeatable host validation - REFERENCE

```
hostnamectl
ip -br addr
ip route
resolvectl status
free -h
lsblk
sudo pvs; sudo vgs; sudo lvs
df -h /
```

Expected: hostname soc-ubuntu01; intended subnet and default route; resolver includes 192.168.150.10; root filesystem reflects the expanded LVM volume. Exact partition/device names were not retained, so do not run blind growpart/lvextend commands without inspecting lsblk and LVM first.

Troubleshooting

• No network: inspect ESXi port group, guest interface, address/mask, gateway, and route; test gateway reachability before changing firewall rules.

• DNS failure: verify resolver and domain DNS reachability; distinguish name-resolution failure from connectivity failure.

• Disk still small: compare lsblk, lvs and df -h; expand only the correct logical volume and filesystem after backup.

4. Phase 2 - Wazuh all-in-one deployment

Recorded state - VERIFIED

• Wazuh Manager 4.14.9, Indexer 4.14.9, Dashboard 4.14.9 and Filebeat 7.10.2 active on soc-ubuntu01.

• Dashboard reachable at https://192.168.150.50. Browser warning expected for a self-signed certificate; validate the certificate fingerprint before trusting it.

Reproduction and validation commands - REFERENCE

```
sudo systemctl status wazuh-manager --no-pager
sudo systemctl status wazuh-indexer --no-pager
sudo systemctl status wazuh-dashboard --no-pager
sudo systemctl status filebeat --no-pager
sudo ss -ltnp | grep -E ':(1514|1515|443|9200)\b'
sudo journalctl -u wazuh-manager -n 50 --no-pager
```

A standard Wazuh 4.14 all-in-one installer reference is shown below for rebuilding a fresh, dedicated VM. This command was not recovered as a verified Lab 08 execution transcript. Confirm current vendor instructions and prerequisites before running; do not rerun on an already configured instance.

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

Troubleshooting

• Dashboard unavailable: check dashboard and indexer services, HTTPS port 443, browser URL, firewall, and logs.

• Agent not connecting: distinguish enrollment port 1515 from ongoing event traffic on 1514; confirm host firewall and routing.

• Certificate warning: expected for self-signed TLS; do not disable certificate validation in production as a permanent workaround.

5. Phase 3 - Windows endpoint enrollment and event ingestion

Verified endpoint state

• IAM-MGMT01, Windows 11 Pro, domain corp.example.test, agent ID 001, Wazuh 4.14.9, group default.

• Windows service WazuhSvc Running and Automatic. Agent log recorded connection to 192.168.150.50:1514/tcp at November 2, 2026 14:49:08.

• Wazuh dashboard Endpoints view showed IAM-MGMT01 active, ID 001, IP 192.168.150.30.

• Custom Windows Application Event ID 101, source SOC-Lab08, successfully forwarded and observed.

Windows PowerShell validation - REFERENCE

```
Get-Service -Name WazuhSvc
Test-NetConnection 192.168.150.50 -Port 1514
Test-NetConnection 192.168.150.50 -Port 1515
Get-WinEvent -FilterHashtable @{
    LogName='Application'; Id=101
} -MaxEvents 10 | Select TimeCreated,ProviderName,Id,Message
```

The Wazuh dashboard/manager should show the enrolled endpoint as active. Network reachability alone is not equivalent to an active agent; confirm the server-side agent heartbeat and actual event ingestion.

Temporary full-archive logging - VERIFIED

For ingestion troubleshooting, manager setting `<logall_json>yes</logall_json>` was temporarily enabled, allowing the Application event to be observed in archives.json. It was subsequently restored to `<logall_json>no</logall_json>`. Avoid leaving verbose archives enabled without a retention/capacity plan.

Troubleshooting

• Agent service running but endpoint inactive: inspect Windows agent log and manager reachability; verify manager address and host firewalls.

• Event absent in Threat Hunting: an ingested event may not trigger an alert rule; confirm archive ingestion separately when appropriate.

• Wrong machine or event source: verify hostname and event provider before exporting evidence.

6. Phase 4 - File integrity monitoring (FIM)

VERIFIED: Real-time monitoring configured for C:\SOC-Lab08. File creation, modification and deletion were observed with the following Wazuh rules.

Activity	Wazuh rule	Level
Create file	554	5
Modify file	550	7
Delete file	553	7

Repeatable local test - REFERENCE

```
$TestDir = 'C:\SOC-Lab08'
New-Item -ItemType Directory -Force -Path $TestDir
$TestFile = Join-Path $TestDir 'fim-validation.txt'
Set-Content -Path $TestFile -Value 'initial test'
Add-Content -Path $TestFile -Value 'modified test'
Remove-Item -Path $TestFile
```

Check Wazuh events for the expected rule IDs and path. Timing can depend on agent configuration and ingestion delay. The original agent configuration XML was not available in the retrieved record, so the exact FIM stanza is not reproduced as verified.

Troubleshooting

• No FIM alert: verify directory path, Windows permissions, Wazuh agent FIM configuration, real-time monitoring flag, and service restart if configuration changed.

• Create seen but modify/delete absent: check alert time range, monitored path, and event filtering before changing rules.

7. Phase 5 - Security configuration and authentication visibility

Security Configuration Assessment - VERIFIED

Benchmark / metric	Result
CIS Windows 11 Enterprise benchmark	Reviewed in Wazuh SCA
Passed checks	125
Failed checks	348
Not applicable	9
Displayed score	26%
Domain minimum password length	7 characters
CIS recommendation noted in lab	14 characters

Decision recorded: Option A - document the password-policy gap without modifying Active Directory. This manual does not direct AD policy remediation as a completed action.

Authentication monitoring - VERIFIED

Windows Security Event ID 4624 (successful logon) and Event ID 4625 (failed logon) were both observed in the monitoring workflow. The controlled failure investigated in detail in Phases 6-7 is Event 4625, Record 46309.

Validation commands - REFERENCE

```
Get-WinEvent -FilterHashtable @{
    LogName='Security'; Id=4624
} -MaxEvents 5 | Select TimeCreated,Id,RecordId
Get-WinEvent -FilterHashtable @{
    LogName='Security'; Id=4625
} -MaxEvents 5 | Select TimeCreated,Id,RecordId
```

Troubleshooting and scope

• A Wazuh SCA finding is a benchmark assessment, not proof of exploitation.

• Do not change domain password policy without a change-control decision; the lab explicitly chose documentation only.

• Event 4740 not found on a workstation does not prove absence of domain-controller account lockouts.

8. Phase 6 - Authorized failed-authentication test

Test setup and observed result - VERIFIED

A temporary local Windows account, SOC-Lab08-Test, was created for the controlled test. A deliberately incorrect password was supplied to a loopback IPC$ connection, resulting in Windows System Error 1326 and a failed network logon. The temporary account was later deleted and its absence confirmed.

Command pattern used - VERIFIED

```
net use \\127.0.0.1\IPC$ /user:.\SOC-Lab08-Test "Lab08-Invalid-Password-2026!"
```

This is a historical lab command containing an intentionally incorrect test password, not a valid secret. Do not repeat against production systems or accounts.

Field	Recorded value
Endpoint / account	IAM-MGMT01 / SOC-Lab08-Test
Event ID / record ID	4625 / 46309
Local event time	Nov 2, 2026 18:11:28
Logon type / package	3 (network) / NTLM
Source IP / port	127.0.0.1 / 61968
Status / substatus	0xC000006D / 0xC000006A
Wazuh alert	Rule 60122, level 5
Wazuh description	Logon Failure - Unknown user or bad password
Dashboard timestamp	Nov 2, 2026 18:11:19.634

Source event verification - VERIFIED

```
Get-WinEvent -LogName Security -FilterXPath   "*[System[EventID=4625 and EventRecordID=46309]]"
```

The event was exported as XML and checked for Computer IAM-MGMT01.corp.example.test, EventID 4625, RecordID 46309, account SOC-Lab08-Test, NTLM, source 127.0.0.1 and the two status codes.

Cleanup - VERIFIED

```
Remove-LocalUser -Name 'SOC-Lab08-Test'
Test-Path 'LocalUser:\SOC-Lab08-Test'
$null -eq (Get-LocalUser -Name 'SOC-Lab08-Test'   -ErrorAction SilentlyContinue)
```

Observed cleanup: Test-Path False; null check True. The event query found exactly one matching 4625 during the 18:00-19:00 window on IAM-MGMT01. No local 4740 event was found in that window.

9. Phase 7 - SOC investigation and response

Incident summary - VERIFIED

Incident LAB08-SOC-003 was assessed as low SOC severity because the authentication failure was deliberately generated during an authorized exercise. The Wazuh rule level was 5; rule level and SOC severity are different fields. No unauthorized access or service disruption was identified within the reviewed endpoint/time-window scope. No emergency containment or recovery was required.

Investigation workflow

• Identify Wazuh alert Rule 60122, level 5, associated with IAM-MGMT01 agent 001.

• Correlate to Windows Event 4625, Record 46309; confirm source computer, account, NTLM, logon type 3 and status codes.

• Check local 4625 events from 18:00 to 19:00; exactly one was found. Query local 4740; none found in that same period.

• Confirm temporary account deletion; classify the event as authorized lab activity.

• Export Wazuh alert CSV and Windows event XML; create timeline, response-action record and SOC incident report.

• Copy the Wazuh CSV to the incident directory and verify source/destination SHA-256 hashes match.

Evidence register - VERIFIED

File / location	Bytes	Verification
Evidence\LAB08-SOC-003-Windows-4625-Verified.xml	1462	XML identifiers and event fields checked
Evidence\LAB08-SOC-003\LAB08-SOC-003-Wazuh-Alert-60122.csv	149	SHA-256 copy match True
Evidence\LAB08-SOC-003\Incident-Timeline.md	1655	Created; first lines checked
Evidence\LAB08-SOC-003\Incident-Response-Actions.md	1996	Created; first lines checked
Evidence\LAB08-SOC-003\SOC-Incident-Report.md	3843	Created; first 20 lines checked

All five paths were confirmed present and nonempty by the final PowerShell inventory: All evidence files present and nonempty: True. This does not independently validate the full contents of each Markdown file.

Evidence copy and SHA-256 verification - VERIFIED

```
$CsvName = 'LAB08-SOC-003-Wazuh-Alert-60122.csv'
$CsvPath = Join-Path $env:USERPROFILE "Downloads\$CsvName"
$IncidentDir = 'C:\SOC-Lab08\Evidence\LAB08-SOC-003'
$Destination = Join-Path $IncidentDir $CsvName
Copy-Item -Path $CsvPath -Destination $Destination -ErrorAction Stop
$OriginalHash = Get-FileHash $CsvPath -Algorithm SHA256
$CopiedHash = Get-FileHash $Destination -Algorithm SHA256
$OriginalHash.Hash -eq $CopiedHash.Hash
# Observed: True
```

Original CSV SHA-256: `2F0FE5D1BCED17242A3993F60F72108C7FFDE99F0A0E2BF860E6585ED4EDB81A`. The copied CSV was verified to have the same hash.

Final inventory command - VERIFIED

```
$EvidenceDir = 'C:\SOC-Lab08\Evidence'
Get-ChildItem -Path $EvidenceDir -File -Recurse |
  Select-Object Name,DirectoryName,Length,LastWriteTime |
  Sort-Object DirectoryName,Name
```

Timeline and closure nuances

• Windows event time (18:11:28) and Wazuh dashboard time (18:11:19.634) differ by about nine seconds. No validated time-zone/clock synchronization reconciliation was recorded; retain each source timestamp as displayed.

• An earlier incorrectly exported XML contained an unrelated IAM-DC01 Event 4672 (Record 463609) but had a misleading 4625 filename. It is excluded from primary incident evidence. The verified IAM-MGMT01 XML is the authoritative event export for this incident.

• The SOC decision was to close as benign authorized activity. The existing report text initially recommended closure after verification; a separate on-disk closure-status update was not shown in the recovered transcript. Do not represent that update as performed.

10. Troubleshooting log and fixes

Issue	Diagnosis / resolution
Initial XML export failed	C:\SOC-Lab08\Evidence did not exist; created directory before export.
Wrong XML event exported	Initial file was IAM-DC01 Event 4672, not target 4625; verified host, exact event ID/record ID and exported corrected XML.
CSV absent from evidence inventory	Original was in Administrator Downloads; copied into incident directory and compared SHA-256 hashes.
PowerShell hash table truncated	Compared hash strings directly; output True.
Wazuh archive ingestion needed	Temporarily enabled logall_json=yes to inspect archives.json, then reverted to no.
Agent troubleshooting	Checked WazuhSvc status, connection log and dashboard active state; network port reachability is not sufficient alone.
Security baseline gap	Minimum password length 7 vs benchmark 14; documented finding; no AD policy changes.

11. Operational recheck and maintenance

SOC Ubuntu - read-only health checks

```
hostnamectl
ip -br addr
sudo systemctl is-active wazuh-manager wazuh-indexer   wazuh-dashboard filebeat
sudo ss -ltnp | grep -E ':(1514|1515|443|9200)\b'
df -h /
```

IAM-MGMT01 - read-only health checks

```
hostname
Get-Service WazuhSvc
Test-NetConnection 192.168.150.50 -Port 1514
Get-ChildItem 'C:\SOC-Lab08\Evidence' -File -Recurse |
  Select Name,Length,FullName
```

For an operational runbook, record date, operator, command, output, ticket/incident reference and any approved changes. Avoid rerunning failed-logon tests unnecessarily after the exercise is complete.

12. Rebuild acceptance checklist

- [ ] SOC VM hostname, address, route, DNS and disk capacity validated
- [ ] Wazuh manager, indexer, dashboard and Filebeat active
- [ ] Dashboard HTTPS accessible and TLS trust reviewed
- [ ] Windows Wazuh agent 001 active and reporting
- [ ] Custom Application Event 101 ingestion confirmed
- [ ] FIM create/modify/delete rules 554/550/553 observed
- [ ] SCA baseline recorded without unauthorized AD changes

• [ ] Windows logon Event 4624 and failed logon Event 4625 observed

• [ ] Authorized 4625 / Rule 60122 correlation verified

• [ ] Temporary test account removed

• [ ] Five incident evidence files present and nonempty

• [ ] CSV source/copy SHA-256 equality verified

• [ ] Incident classified low severity and closure decision recorded

13. Evidence handling and publication guidance

The report describes an internal lab using private IP addresses and local machine names. For public portfolio publication, redact user identifiers, secrets, agent enrollment keys, certificates, session tokens and sensitive event fields. Preserve original evidence privately. Publish sanitized screenshots, architecture, project narrative, methodology, results, and selected safe command examples.

Suggested portfolio summary: Deployed a Wazuh SOC stack on Ubuntu in VMware ESXi, enrolled a Windows endpoint, validated Windows Event Log collection and FIM, reviewed CIS posture, and completed an authorized failed-authentication investigation with event correlation, response documentation and SHA-256 evidence verification.

14. Record completeness and limitations

• Original screenshots and exact installation transcripts for every Phase 1-2 command were not attached to this PDF generation request. The document provides reference commands where needed and marks their provenance.

• The precise original Windows agent installer/enrollment command and complete Wazuh XML configuration were not available; they are not fabricated.

• The full bodies of the three Markdown reports were not accessible from the Windows endpoint; the manual summarizes confirmed excerpts and reported purpose, not a full independent audit.

• Evidence integrity confirmation was specific to the Wazuh CSV copy. Other files were checked for existence, nonzero size and selected contents, not individually hashed.

• Incident conclusions are limited to the authorized lab endpoint and time window; no enterprise-wide or domain-wide compromise assessment was conducted.

Document status: Finalized as a reconstructed, evidence-labeled lab runbook from the verified conversation record. The report does not assert that every reference command was executed.